The effects of solar radiation on daily and seasonal stem increment of canopy trees in European temperate old-growth forests

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Summary

- It is well established that solar irradiance greatly influences tree metabolism and growth through photosynthesis, but its effects acting through individual climate metrics have not yet been well quantified. Understanding these effects is crucial for assessing the impacts of climate change on forest ecosystems.
- To describe the effects of solar irradiance on tree growth, we installed 110 automatic dendrometers in two old-growth mountain forest reserves in Central Europe, performed detailed terrestrial and aerial laser scanning to obtain precise tree profiles, and used these to simulate the sum of solar irradiance received by each tree on a daily basis. Generalized linear mixed-effect models were applied to simulate the probability of growth and the growth intensity over seven growing seasons.
- Our results demonstrated various contrasting effects of solar irradiance on the growth of canopy trees. On the one hand, the highest daily growth rates corresponded with the highest solar irradiance potentials (i.e. the longest photoperiod). Intense solar irradiance significantly decreased tree growth, through an increase in the vapor pressure deficit. These effects were consistent for all species but had different magnitude.
- Tree growth is the most effective on long rainy/cloudy days with low solar irradiance.

Introduction

The sun provides energy to power photosynthesis, which drives the production process of assimilates that in trees is used in the formation of new wood tissue (Pallardy, 2008; Hartmann & Trumbore, 2016). In the temperate zone, the highest rates of tree growth are synchronized with the photoperiod (Rossi *et al.*, 2016; Huang *et al.*, 2020). However, intense solar irradiance increases evapotranspiration and can cause drought stress or foliage injuries (Urban *et al.*, 2007; Hartmann & Trumbore, 2016; Still *et al.*, 2023). To minimize such damage, plants have evolved photoprotective mechanisms that inhibit metabolic processes, known as photoinhibition (Murchie *et al.*, 1999; Kothari *et al.*, 2021). However, different metabolic capacities and physiological predispositions of individual tree species, or trees of different sizes, may lead to differing responses to the same intensity of solar irradiance (Feng & Dietze, 2013; Vinod *et al.*, 2023). The most pronounced impacts of climate change include rising temperatures and elevated risks of drought (Spinoni *et al.*, 2018; Lewis *et al.*, 2019), which might be expected to result in reduced cloud cover, and increases in the amount of received solar irradiance (Palmer *et al.*, 2004; Sanchez-Lorenzo *et al.*, 2017). Hence, understanding the effects of different aspects of solar irradiance, the total amount of received solar irradiance and the intensity of solar irradiance (the ratio between the potential and received solar irradiance), is important in assessing the likely effects of climate change on forests in the future.

The total amount of solar irradiance received by a tree crown depends on the solar azimuth and zenith angles, leading to significant differences on daily and seasonal scales. Together with the surrounding topography, weather conditions and forest structures, this results in a high spatiotemporal irregularity of solar irradiance within a forest. The position of a tree within a forest stand then determines the overall potential of received solar irradiance. At the same time, solar irradiance influences various meteorological metrics, such as air temperature and the vapor pressure deficit – VPD (Kothari *et al.*, 2021). Temperature, VPD and soil moisture have all been identified as factors affecting turgor, the threshold of which must be exceeded to continue the process of cell expansion (Zweifel *et al.*, 2016; Fatichi *et al.*, 2019; Cabon *et al.*, 2020b; Peters *et al.*, 2021). In addition, intense solar irradiance increases leaf temperatures (Still *et al.*, 2022), reducing carbon assimilation (Crous *et al.*, 2023). Beyond a certain threshold, the amount of received solar irradiance may surpass a tree's photosynthetic capacity (Adams *et al.*, 2013), leading to photoinhibition. On the other hand, the highest tree growth corresponds to the days with the longest photoperiod (Huang *et al.*, 2020). These contradictory effects of solar irradiance on tree growth have yet to be adequately addressed.

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Long-lasting drought periods can lead to stem water deficits (Salomón *et al.*, 2022), reducing tree growth (Cabon *et al.*, 2020a; Liu *et al.*, 2020), and inducing large-scale mortality (Meir *et al.*, 2015), either directly or by enhancing susceptibility to bark beetle attacks (Netherer *et al.*, 2024). The effects of climate and drought on tree growth are also affected by different stages of tree ontogeny across a population (Trouillier *et al.*, 2019; Anderson-Teixeira *et al.*, 2022). Similarly, species-specific effects of climate and drought have been identified due to differences in tree morphology, rooting depth and wood anatomy (Eilmann & Rigling, 2012; Kašpar *et al.*, 2021). However, if soil moisture was sufficient for tree growth, VPD may become more important (Tumajer *et al.*, 2022).

Species-specific responses of tree growth to climate metrics have already been observed at the cellular level – during xylogenesis (Rossi *et al.*, 2016; Chen *et al.*, 2019), with results suggesting a significant influence of climate conditions on the timing of wood formation (Treml *et al.*, 2015). Usual time interval of sampling xylogenesis samples is ≥ 5 d (Rossi *et al.*, 2007), which is substantially longer than the potential incorporation of assimilates into new tissue (Feng & Dietze, 2013), making the relative importance of individual (and very dynamic) meteorological metrics hard to differentiate. Automatic dendrometers can capture differences in stem diameter at a narrower time scales, and can thus enable linking of meteorological data with tree growth (cell differentiation and expansion) at daily or even sub-daily scales (Zweifel *et al.*, 2016, 2021; van der Maaten *et al.*, 2018; Ježík *et al.*, 2021; Etzold *et al.*, 2022; Tumajer *et al.*, 2022).

In this study we aimed to untangle the effects of solar irradiance and VPD on the tree growth dynamics in a (sub)montane beech-dominated forests. Our specific goal was to quantify the effects of different aspects of solar irradiance, specifically relative daily irradiation (RDR, expressing the intensity of solar irradiance in percent of the daily potential), specific daily solar irradiance (SDSI, capturing the amount of solar radiance received by an individual tree in Joules per 1 m^2 per 1 h of the sun-exposed tree crown) and the photoperiod (day length). Our study was based on the daily (and seasonal) stem increments of canopy trees of three important European tree species – *Abies alba* Mill., *Fagus sylvatica* L., and *Picea abies* (L.) Karst. Since solar radiation is a key driver of most meteorological metrics (e.g. temperatures), we hypothesized a unimodal effect on tree growth, with both low and high solar irradiance significantly limiting tree growth (Noyer *et al.*, 2023; Still *et al.*, 2023). This hypothesis is seemingly in contrast with the expectation of high growth rates during days with the longest photoperiods (Rossi *et al.*, 2016; Huang *et al.*, 2020). Furthermore, we expected that conifer and broad-leaved species would respond differently to solar irradiance.

Materials and Methods

Study sites

The study was conducted in two of the oldest forest reserves in Central Europe, the Boubín and Žofín old-growth forests (Supporting Information Fig. S1). Both sites are situated in mountainous areas in southern Bohemia near the Czech-German and Czech-Austrian borders, and are strictly protected without (or with very limited) direct human influence for > 150 yr. The long-term dominance of naturally driven processes has resulted in diverse vertical and horizontal forest structures (Král *et al.*, 2014, 2018).

The Boubín old-growth forest (48°58′N, 13°48′E) is located at elevations between 925 and 1105 m above sea level (asl), and has an almost equal presence of *P. abies* and *F. sylvatica*. The Žofín old-growth forest (48°40′N, 14°42′E) is located at elevations from 730 to 837 m asl, with *F. sylvatica* dominating the tree species with the addition of *A. alba* on dryer soils and *P. abies* on hydromorphic soils (Kašpar *et al.*, 2021). The mean annual temperature and precipitation totals in the 2016–2022 period were 6.2°C and 871 mm in Boubín and 7.1°C and 823 mm in Žofín, respectively (E-OBS; Cornes *et al.*, 2018). The mean growing season SPEI ranged from −0.25 to −0.7 in Žofín (2016–2022) and from −0.23 to −0.06 in Boubín (2019–2022).

Dendrometer measurements and data preparation

For this study we collected dendrometer data during four growing seasons (2019–2022) in Boubín and seven growing seasons (2016–2022) in Žofín. Automatic band dendrometers (DLR26c; EMS Brno, Brno, Czechia; Fig. 1a) were installed on 110 canopy trees (size parameters of the studied trees are shown in Fig. S2). The dendrometers continuously measured (in 1-h intervals) the circumference of the tree stem with a precision of < 1 μm. All dendrometers were first installed in Žofín in the beginning of the 2016 growing season. At the end of the 2018 growing season, 19 devices were relocated to Boubín (Table S1).

Stem circumference were recalculated into stem diameter data in 1-h intervals (Fig. 1b), allowing us to calculate the daily stem diameter maxima of each studied tree (van der Maaten *et al.*, 2018). However, daily stem diameter maxima still contained fluctuations due to the stem cycle (the continuous process of stem shrinkage and stem expansion related to the stem water content). To remove this fluctuation, we used the zero-growth approach (Zweifel *et al.*, 2016; Knüsel *et al.*, 2021). The calculation of zero-growth, is based on a running maximum over the continuous stem data, where only the next maximal value is considered as growth increment and days with such an increment are considered as 'growing days'. The other days are then considered as a nongrowing days. We further divided nongrowing days into days with stem shrinkage and days with stem tissue refiling (Fig. 1c,d). In order to remove the 'noise' due to the rehydration/sequentially ceasing growth rate of the stem at the beginning and end of the growing season, the upper and lower 2.5% of the radial increment data were omitted. All following analyses are thus based on a continuous period representing a 95% of the annual increment (Zweifel *et al.*, 2016; Knüsel *et al.*, 2021). Using a 90% annual increment threshold did not significantly affect the results (data not shown). Finally, data for each tree in each growing season were standardized to obtain a percentage of the cumulative stem growth on a scale of 0–1. This was done in

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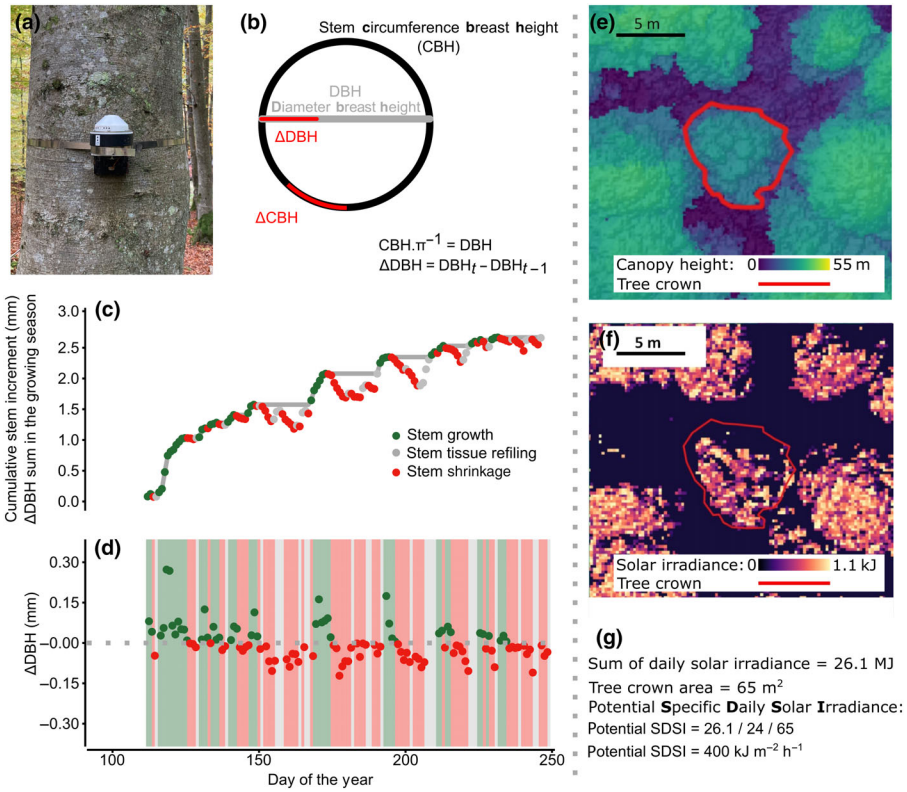

Fig. 1 Methodological approach: (a) dendrometer data collection, (b) conversion to continuous stem diameter data, (c) determination of days of growth, days of stem shrinkage, and the stem tissue refiling of cumulative growth (the gray line represents the zero-growth value), (d) calculation of daily increments and daily shrinkage, (e) identification of the tree crown of one studied tree, (f) simulated daily solar irradiance on 1 d, (g) calculation of the potential specific daily solar irradiance in 1 d.

order to enable a comparison of growth timing and daily stem increments for trees of different dimensions.

Meteorological data

For both sites we extracted daily meteorological observations for the period 1950–2022 from the Ensembles daily gridded OBServational dataset (E-OBS) at a resolution of 0.1° (Cornes *et al.*, 2018). Specifically, we focused on daily data of: mean temperatures (°C), precipitation totals (mm d^{-1}), mean solar irradiance (W m^{-2}) and mean relative air humidity (%). In Žofín, a meteorological station was established in 2016, measuring temperature, relative humidity, precipitation and solar irradiance at 10-min intervals. The station is located in an edaphic gap (an open area without shading by canopy trees throughout the growing season). The temperature and solar irradiance were measured at 2 m and precipitation at 1 m aboveground. From those values we calculated the same metrics as those extracted from the E-OBS database.

Due to the similar seasonality, we then standardized the temperature and solar irradiance data. To do so, for temperature datasets we calculated mean daily temperatures for the period 1950–2020 (the full range of the E-OBS dataset used) and subsequently calculated anomalies from the long-term mean values (Fig. S3A,B). Due to the high correlation of E-OBS and *in situ* data (Fig. S4), for subsequent calculations, we used data from the E-OBS database for Boubín and data from the local meteorological station for Žofín. This is because meteorological station in Boubín was absent from 2019 to 2022. For the solar irradiance dataset, we calculated potential daily irradiance from running maxima in the long-term record (Fig. S3C,D). Dividing the actual daily irradiance by the potential daily irradiance in corresponding days we obtained the Relative Daily irRadiation rate (RDR) on a scale from 0 to 1 (Fig. S3C,D). Similarly, we calculated temperature anomalies (K) by first calculating long-term daily mean temperature from the E-OBS dataset and then calculating anomalies from long-term values. In all variables, except precipitation totals, we use daily means. For precipitation, we calculated two days sums (2-d) for two consecutive days of the year (DOY; $precipitation_{DOY}$ and $precipitation_{DOY-1}$). The mean daily temperatures and mean daily relative humidity were used to calculate the vapor pressure deficit (VPD) according to Howell & Dusek (1995).

The data of volumetric soil water content (SWC) at the depth 30 cm were extracted from ERA5 land database (Hersbach *et al.*, 2023), and used to calculate the relative extractable water (REW; a normalized metric of soil water arability) according to Flo *et al.* (2021). We also conducted *in situ* measurements of the soil water potential (SWP) using Delmhorst GB1/GB2 sensors and MicroLog SP3 dataloggers (EMS Brno). Sensors were placed at both sites during the middle of the growing season 2020 at depths of 10 and 30 cm. In total five devices were placed in Boubín and six in Žofín. Daily mean values of SWP were calculated and then bootstrapped to get 95% confidence interval of the mean daily SWP for each site.

Tree crown data and the quantification of solar irradiance

Aerial laser scanning (UAV) was performed in Žofín in April 2018, with an average density exceeding 4000 points per 1 m^2

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(Kellner *et al.*, 2019). Boubín was scanned in November 2019 with an average density of *c.* 3000 points per 1 m^2. Digital surface models with 25 cm resolution were generated for both sites, and used to identify and vectorize sun-exposed parts of the crowns of the studied trees (considering a UAV-based raster of tree crowns). Both datasets were used to simulate the solar irradiance of each sun-exposed tree crown under study at a 1-h resolution. Aggregating the data by DOY, we calculated the potential daily irradiance of each tree crown of studied trees. This simulation was done using the R package INSOL (Corripio, 2021; R Core Team, 2023). As a climate predictor necessary for simulation, the long-term mean daily data extracted from the E-OBS database for the entire available database period were used. An example of the script used can be found in Methods S1. First, we calculated the potential solar irradiance for each hour in the year (in total 8760 simulations per tree). Then we calculated mean daily solar irradiance for each tree (in joules per hour). This value was further divided by the area of the visible tree crown of each tree. Calculating the mean specific potential daily solar irradiance of 1 m^2 of each tree crown per 1 h we get an energy received by tree in J m^{-2} h^{-1}. Consequently, using the potential daily solar irradiance, RDR and area of sun-exposed parts, we calculated the Specific Daily Solar Irradiance (SDSI), for each tree crown. Using the INSOL R package we calculated the day length for both sites (Corripio, 2021).

Statistical analysis

Differences between stem increments among individual tree species at both sites (both annual and mean daily), the length of the growing season and the number of growing days were assessed using the Kruskal–Wallis test. The asymmetry in the distribution of the growing days (days with identified stem increment) and the growing season days (all days since the beginning to the end of the growing season) with respect to the DOY, SDSI, RDR and VPD, were all evaluated using the Kolmogorov–Smirnov test. Differences in individual climate metrics between growing and nongrowing days were then evaluated using the Kruskal–Wallis test. We used this test instead of ANOVA due to the different size of the datasets of growing days, days with stem shrinkage and days with stem tissue refiling.

The probability of a particular day being a growing day was modelled using generalized mixed-effects models with a binomial distribution, using R package LME4 (Bates *et al.*, 2015), where growing days had a value of 1 and nongrowing days a value of 0. The growth intensity was calculated using mixed-effect models. For each variant, we calculated two pairs of models (M1 and M2). In both models, the tree and metrics of solar irradiance (SDSI and RDR) were treated as random effects: the tree was considered as the random intercept, and solar irradiance as the random slope. The day length, REW, VPD, temperature anomalies, precipitation, SDSI and interaction of SDSI and VPD were fixed predictors in M1. In M2, RDR was used instead of SDSI. The full formula is given in Table S2. All predictors were scaled to obtain unit variance and zero mean in order to reduce the collinearity among predictors in the models (VIF < 4). The coefficient of determination was calculated using MUMIN R package (Bartoń, 2016).

The mean growth curve of each species was calculated from bootstrapped stem increments. Classes for bootstraping were set at 10 kJ m^{-2} h^{-1} in the case of SDSI and 0.01 (1%) in the case of RDR. In the end bootstrapped data were fitted by a smooth spline. The effect of each metric was estimated by linear regression and confirmed by a linear mixed-effect model of the same structure as the model simulating the growth probability (Bates *et al.*, 2015). Owing to the observed mutual correlation between VPD and the metrics of solar irradiance, we conducted a mediation analysis (Imai *et al.*, 2010), included in the MEDIATION R package (Tingley *et al.*, 2019), using linear regression to predict VPD using either RDR or SDSI. This approach enabled us to detach the comprehensive impact of the solar irradiance metrics, in conjunction with the effect of solar irradiance on VPD. Subsequently, the interaction between the solar irradiance metrics and VPD was evaluated in linear mixed-effects models. All statistical analysis were performed in R (R Core Team, 2023).

Results

We used generalized linear mixed-effect models to simulate (1) the probability of growth on particular days, and (2) the daily radial increment of trees depending on selected meteorological variables. The model estimating the probability of growing days showed the highest positive effect of day length and precipitation ($P < 0.05$; Fig. 2a–e). The significantly ($P < 0.05$) negative effect of VPD was usually the third highest (Fig. 2a–e). On the contrary, both SDSI and RDR metrics had lower effect than VPD and day length and were mostly statistically insignificant (Table 1). The *R*-squares of the models ranged from 0.35 to 0.47 (Table 1). Still, there existed a significant difference ($P < 0.05$) in the distribution of both nonsignificant (RDR and SDSI) and significant (VPD and day length) predictors on growing and nongrowing days (Fig. S5).

The model of radial increment showed that growth rates increased with daylight and decreased with increasing RDR/SDSI and VPD (Fig. 3; Table 2). The positive effect of day length ranged from 0.21% to 0.34% of the annual increment per 1 h of daylight, with > 50% of growth realized on days with > 15.5 h of daylight (in total 50–65% of growing season days). The daily stem increment decreased by 0.10–0.27% per increase of VPD by 0.1 kPa (Fig. 3c,d). *Picea abies* realized most of its growth on days with significantly lower VPD than *F. sylvatica* (Fig. 3c,d). The daily stem increment decreased by 0.01–0.1% per increase of specific solar irradiance by 10 kJ m^{-2} h^{-1} (Fig. 3e,f) and by 0.01–0.04% per increase of RDR by 0.01 (1%; Fig. 3g,h). Considering both metrics of solar irradiance, *F. sylvatica* was more tolerant to solar irradiance while *P. abies* was the most sensitive (Fig. 3e–h).

In the linear mixed-effects model, the effects of day length, VPD, and both metrics of solar irradiance were found to be statistically significant ($P < 0.05$) and consistent with data presented in Fig. 3 (Fig. 4; Table 2). When VPD was mediated, the impact of both SDSI and RDR exceeded the direct,

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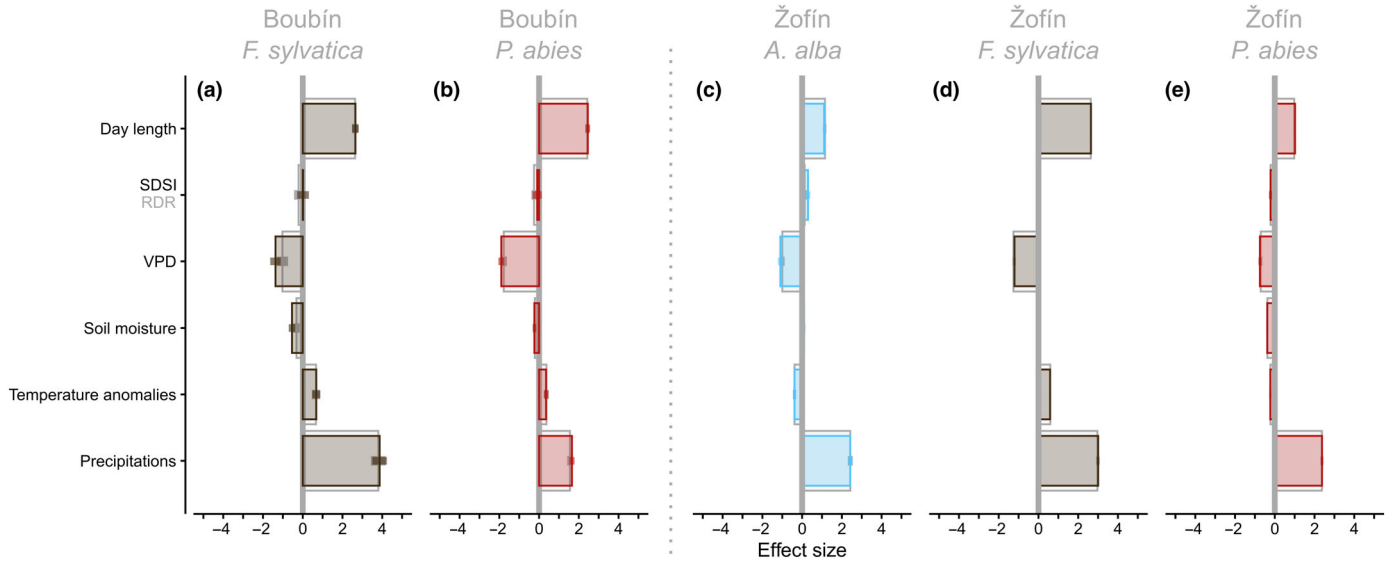

Fig. 2 Effect sizes on the probability of daily growth. (a) *Fagus sylvatica* in Boubín; (b) *Picea abies* in Boubín; (c) *Abies alba* in Žofín; (d) *F. sylvatica* in Žofín; and (e) *P. abies* in Žofín. Individual species are marked in color. Colored bars denote the model with specific daily solar irradiance (SDSI) as a metric of solar radiation; gray boxes in the background denote the second model with the relative daily irradiation (RDR) as a predictor of solar radiation. VPD, vapor pressure deficit.

Table 1 Estimates of variables used to predict growth probability using a generalized mixed-effects model.

	Boubín *Fagus sylvatica*		Boubín *Picea abies*		Žofín *Abies alba*		Žofín *Fagus sylvatica*		Žofín *Picea abies*	
	M1	M2	M1	M2	M1	M2	M1	M2	M1	M2
Model variant										
Variance explained (marginal R^2)	0.77	0.77	0.73	0.66	0.73	0.73	0.81	0.81	0.72	0.72
Variance explained by fixed effects (conditional R^2)	0.68	0.68	0.49	0.59	0.61	0.62	0.65	0.68	0.66	0.67
Variance explained by random effects	0.09	0.09	0.23	0.07	0.12	0.11	0.16	0.13	0.06	0.05
Model estimates										
Day length	**1.06**	**1.02**	**0.97**	**1.09**	**0.52**	**0.56**	**1.18**	**1.23**	**0.47**	**0.44**
REW	**−0.41**	**−0.43**	**−0.35**	**−0.33**	**−0.11**	**−0.1**	−0.03	**−0.07**	**−0.39**	**−0.42**
VPD	**−0.53**	**−0.71**	**−1.13**	**−1.18**	**−0.89**	**−0.86**	**−0.8**	**−1.08**	**−0.47**	**−0.73**
Temperature anomalies	**0.33**	**0.35**	**0.23**	**0.21**	*−0.09*	*−0.08*	**0.45**	**0.47**	−0.03	0.00
Precipitation	**2.74**	**2.77**	**1.6**	**1.65**	**2.23**	**2.24**	**2.83**	**2.71**	**2.43**	**2.34**
SDSI/RDR	−0.27	0.17	*1.12*	0.15	0.2	0.15	0.15	0.15	**−0.34**	**−0.11**
SDSI/RDR : VPD	−0.02	0.05	**0.16**	**0.3**	**0.26**	**0.28**	**0.14**	**0.37**	−0.03	**0.26**

Models differing in the solar irradiance metric used were used to predict the data; specific specific daily solar irradiance (SDSI) was used in M1 and relative daily irradiation (RDR) in M2. The coefficient of determination is labeled as R^2, calculation of R^2 was done using MUMIN R package and theoretical R^2 are reported. Statistically significant predictors are bold ($P < 0.05$) or italic ($P < 0.1$). REW, relative extractable water; VPD, vapor pressure deficit.

(unmediated) effect of VPD (Fig. 4; Table 2). Additionally, it is important to highlight the significant influence of precipitation on stem increment, which was the highest across all species. When analyzing the total effect of the RDR and SDSI, the size of the effect of precipitation and solar irradiance appeared similar, albeit with opposing directions. The negative effect of RDR and SDSI increased with increasing tree crown area exposed to solar irradiance (Fig. S6).

The studied trees grew from the beginning of May to the end of August. Conifers (*A. alba* and *P. abies*) at both sites had significantly ($P < 0.05$) longer growing season (129 d in Žofín and 125 d in Boubín), compared with *F. sylvatica* (111 d in Žofín and 107 d in Boubín; Fig. 5a). The temperature at the start of the growing season was significantly different ($P < 0.05$) between the conifers and *F. sylvatica* (7.1 ± 4.4°C and 10.4 ± 4.7°C). The annual increment of *F. sylvatica* and *P. abies* did not differ at either sites ($P > 0.05$), but *A. alba* had significantly ($P < 0.05$) higher annual increments than *F. sylvatica* and *P. abies*. When comparing the annual increment of *P. abies* and *F. sylvatica* between the sites, the growth in Boubín was significantly ($P < 0.05$) higher than in Žofín for both species (Fig. 5b). Growing days accounted for 59.6 ± 22.5% of total days. In Boubín, trees showed a higher number ($P < 0.05$) of growing days than their counterparts in Žofín in all growing seasons

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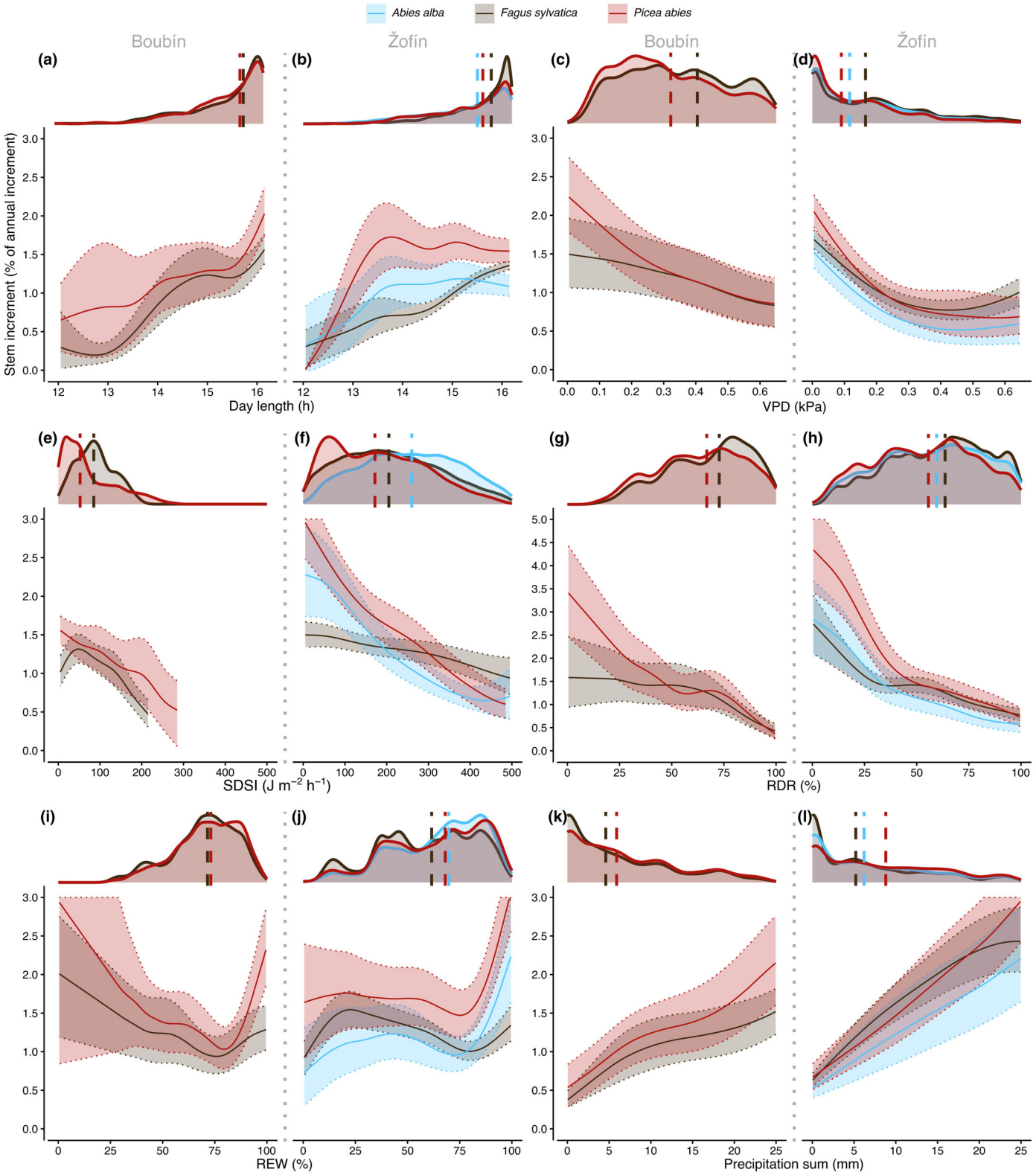

Fig. 3 Effect of selected metrics on the intensity of daily growth. (a, b) Day length in Boubín and Žofín; (c, d) vapor pressure deficit (VPD) in Boubín and Žofín; (e, f) specific daily solar irradiance (SDSI) in Boubín and Žofín; (g, h) the relative daily irradiation (RDR) in Boubín and Žofín; (i, j) the relative extractable water (REW); and (k, l) two days precipitation sums. Above each line graph, the distribution of each indicator is plotted with vertical lines indicating the median value of each species. The solid line represents the bootstrap values of daily stem increments fitted by spline (with 10 degrees of freedom). The transparent areas bounded by a dotted line represent the 95% confidence interval of mean daily stem increments. The individual species are color-coded. REW, relative extractable water.

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Table 2 Estimates of variables used to predict the daily increments using a linear mixed-effects model.

	Boubín *Fagus sylvatica*		Boubín *Picea abies*		Žofín *Abies alba*		Žofín *Fagus sylvatica*		Žofín *Picea abies*	
	M1	M2	M1	M2	M1	M2	M1	M2	M1	M2
Model variant										
Variance explained (marginal R^2)	0.39	0.35	0.42	0.37	0.42	0.43	0.33	0.3	0.45	0.44
Variance explained by fixed effects (conditional R^2)	0.33	0.33	0.36	0.35	0.38	0.37	0.26	0.27	0.37	0.38
Variance explained by random effects	0.07	0.02	0.06	0.02	0.04	0.06	0.08	0.04	0.08	0.44
Model estimates										
Day length	**0.24**	**0.19**	**0.22**	**0.20**	**0.18**	**0.14**	**0.22**	**0.19**	**0.15**	**0.11**
REW	**−0.30**	**−0.30**	**−0.28**	**−0.27**	**−0.22**	**−0.22**	**−0.26**	**−0.25**	**−0.26**	**−0.27**
VPD	**−0.28**	**−0.27**	**−0.28**	**−0.24**	**−0.23**	**−0.27**	**−0.21**	**−0.22**	**−0.16**	**−0.28**
Temperature anomalies	**0.25**	**0.25**	**0.23**	**0.25**	**0.19**	**0.2**	**0.29**	**0.29**	**0.15**	**0.16**
Precipitation	**0.39**	**0.39**	**0.44**	**0.42**	**0.41**	**0.40**	**0.32**	**0.31**	**0.42**	**0.40**
SDSI/RDR	**−0.24**	**−0.17**	**−0.25**	**−0.19**	**−0.23**	**−0.19**	**−0.20**	**−0.16**	**−0.34**	**−0.23**
SDSI/RDR : VPD	0.00	0.00	**0.02**	0.01	**0.10**	**0.13**	**0.02**	**0.03**	**0.09**	**0.16**
Mediation analysis										
VPD ~ SDSI/RDR	0.53	0.54	0.57	0.49	0.47	0.57	0.51	0.55	0.27	0.51
SDSI/RDR mediated	−0.27	−0.19	−0.32	−0.17	−0.15	−0.15	−0.19	−0.15	−0.10	−0.15
SDSI/RDR direct	−0.24	−0.16	−0.24	−0.17	−0.16	−0.11	−0.18	−0.13	−0.27	−0.14
SDSI/RDR total	−0.51	−0.35	−0.56	−0.34	−0.31	−0.27	−0.36	−0.28	−0.37	−0.29

Models differing in the solar irradiance metric used were used to predict the data; specific daily solar irradiance (SDSI) was used in M1 and relative daily irradiation (RDR) in M2. The coefficient of determination is labeled as R^2, calculation of R^2 was done using MUMIN R package and theoretical R^2 are reported. Statistically significant predictors are bold ($P < 0.05$). REW, relative extractable water; VPD, vapor pressure deficit.

(Fig. 5c; Table S1). *Picea abies* at both sites showed a lower number of growing days than *F. sylvatica* ($P < 0.05$; Fig. 5c). The mean number of days with recorded stem shrinkage was 28.5 ± 14.1%, with opposite species-specific differences in the number of growing days (Fig. 5c). The mean daily increment was 1.3 ± 1.6% and mean daily shrinkage 0.6 ± 0.9% of the annual stem increment (considering only growing days and days with stem shrinkage, respectively). *Picea abies* in Žofín also had the highest daily increments and the highest stem shrinkage (Fig. 5d,e).

Discussion

The effects of solar irradiance on the growth of canopy trees

Although neither RDR nor SDSI had consistently significant effect on the probability of growth on a particular day (Table 1), both had significant negative effects on daily stem increments (Table 2). Indeed, in all models, the effect of VPD was higher that the direct effects of SDSI and RDR on the probability of tree growth, supporting VPD as the dominant factor limiting tree growth (Peters *et al.*, 2021; Etzold *et al.*, 2022). The mediation analysis, however, showed a greater effect of RDR and SDSI on the stem increment (Table 2). Increasing solar irradiance increases leaf temperature over the ambient air temperature (Still *et al.*, 2022; Crous *et al.*, 2023) and increases transpiration requirements (Urban *et al.*, 2007). Higher air temperature enhances the capacity of the air to contain vaporized water and facilitates the advection of water vapor to the upper troposphere, ultimately increasing near surface VPD (Fig. S6; Liu *et al.*, 2004). Subsequently, higher VPD lowers turgor, thereby limiting the physiological processes of cell expansion (Fatichi *et al.*, 2019; Cabon *et al.*, 2020b; Peters *et al.*, 2021).

Most of the growing days occurred on days when RDR was far below the potential solar irradiance, suggesting a negative effect of high direct solar irradiance. Photosynthesis remains sufficiently effective under conditions of sub-light saturation (Körner, 2015), highlighting the significant role of diffuse and reflected light (Gu *et al.*, 2002; Urban *et al.*, 2007). Conversely, an increased quantity of direct solar irradiance can lead to growth reduction due to photoinhibition (Murchie *et al.*, 1999; Adams *et al.*, 2013). This is further corroborated by the changing slope of RDR and SDSI with increasing tree crown area (Fig. S6). In smaller (more shaded) trees, only the top of the crown is irradiated directly, while an increasing tree crown area (higher crown exposure), increases the proportion of the crown receiving direct irradiation (both from the top and the sides). On an increasing gradient of tree crown exposure, photoinhibition of exposed trees is more apparent (Sofo *et al.*, 2009) and the availability of solar irradiance changes from being a source to a stressor (Gessler & Zweifel, 2024). Gradual decrees of stem increment with increasing SDSI thus contradicts our hypothesized unimodal pattern (Fig. 3c,d), while following an almost linear trend. This indicates that for tree growth only a relatively small amount of solar irradiance is necessary (Körner, 2015). On days with high solar radiation, the time available for cell (stem) expansion is strongly restricted (Peters *et al.*, 2023) or limited only to the night hours (Zweifel *et al.*, 2021). By contrast, VPD follows a negative exponential, suggesting a threshold value that defines the possibility of growth (Lockhart, 1965; Zweifel *et al.*, 2021; Etzold *et al.*, 2022).

Consistently with previous studies conducted in temperate forests, most of the growing days of all trees in our study were synchronized with day length (Figs 2a–e, S5P–T; Rossi *et al.*, 2016;

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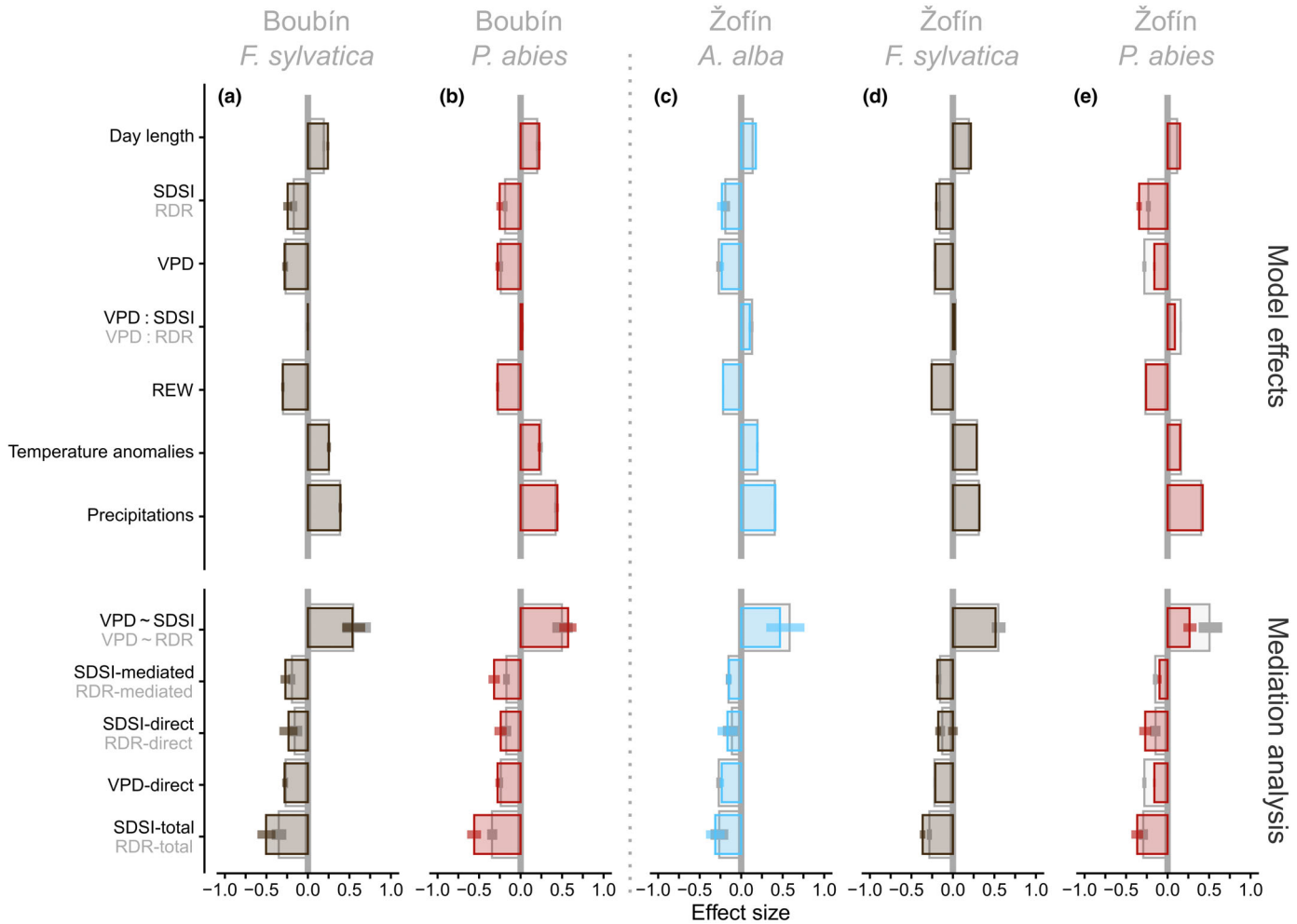

Fig. 4 Effect sizes on daily stem increments: (a) *Fagus sylvatica* in Boubín; (b) *Picea abies* in Boubín; (c) *Abies alba* in Žofín; (d) *F. sylvatica* in Žofín; and (e) *P. abies* in Žofín. Individual species are marked in color. Colored bars denote the model with specific daily solar irradiance (SDSI) as a metric of solar radiation; gray boxes in the background denote the second model with the relative daily irradiation (RDR) as a predictor of solar radiation. REW, relative extractable water; VPD, vapor pressure deficit.

van der Maaten *et al.*, 2018; Huang *et al.*, 2020; Ježík *et al.*, 2021; Zweifel *et al.*, 2021). During the month surrounding the summer solstice, about one-third of the annual stem increment was attained (Fig. S7A–E). Thus, daylength was the only metric with a positive effect connected with solar irradiance. Since the effect of RDR and SDSI on the growth of canopy trees was negative, it cannot be said that this was a positive effect of solar irradiance, but rather a 'side effect' of: (1) generally favorable temperature and precipitation patterns at the beginning of the growing season compared with summer and autumn conditions (Holtanová *et al.*, 2022); and (2) a possibly longer period of time when favorable conditions (intermediate solar radiation) were present.

We identified a strong effect of precipitation on both the probability of growth and growth intensity. On days with precipitation (> 0.1), VPD, RDR, and SDSI were significantly ($P < 0.05$) lower compared to days without precipitation (data not shown). Logically, these conditions align with the previously discussed effects of VPD, RDR, and SDSI. The much greater effect of precipitation in our study than that reported by Etzold *et al.* (2022) might be attributed to our usage of 2-d precipitation totals.

Our results showed sufficient SWP and SWC for tree growth throughout most of the growing season (Fig. S8). Moreover, most of the studied *P. abies* in Žofín grew on hydromorphic soils with sufficient water saturation throughout the growing season, but they showed the fewest number of growing days (Fig. 5c). Our results thus confirmed that for the growth of well hydrated trees, atmospheric conditions are more important driver than soil water supplies (Tumajer *et al.*, 2022). Despite favorable soil moisture a large number of days with intense solar irradiance can significantly reduce periods with suitable conditions for cell enlargement (Cabon *et al.*, 2020b).

Species-specific differences

The growth response of individual tree species to solar irradiance showed similar effect across species, but with differing magnitude

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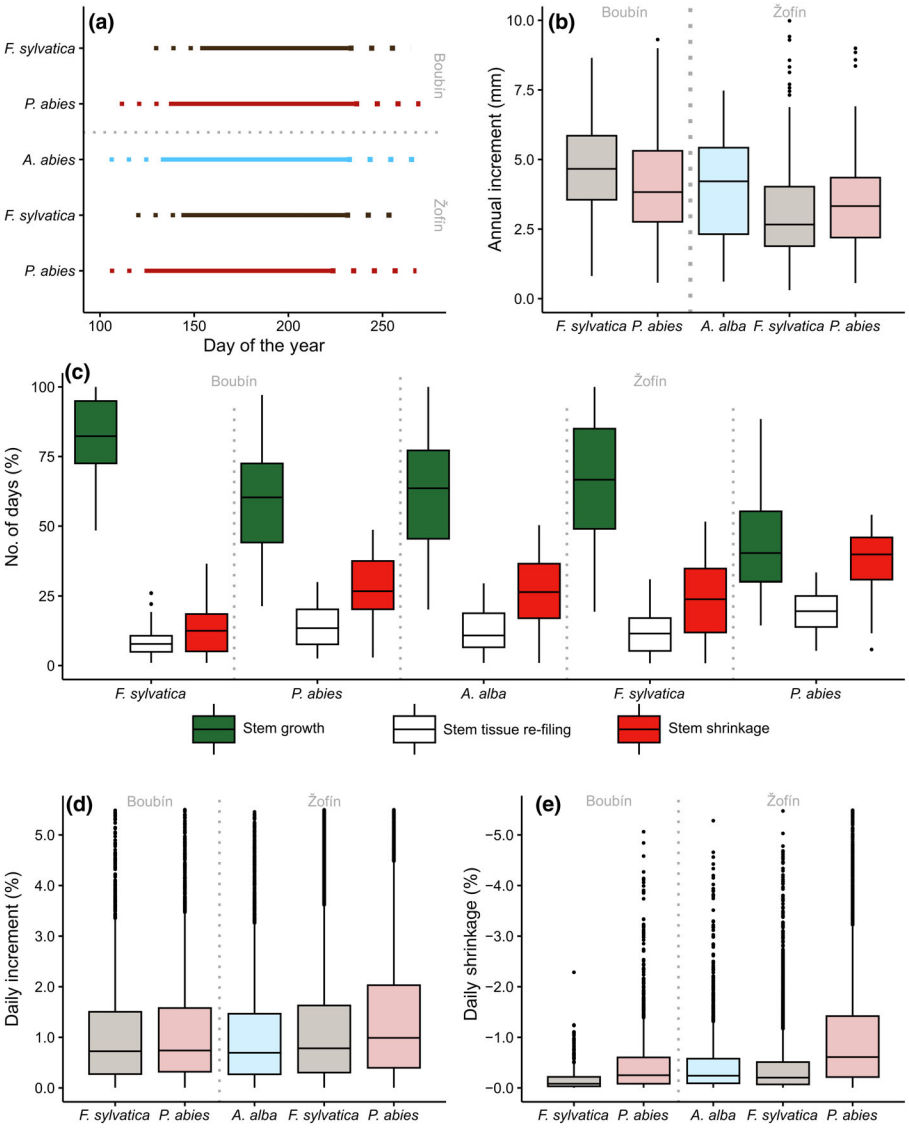

Fig. 5 Basic characteristics of tree growth recorded with dendrometers. (a) Length of the growing season, that is the period in which 95% of growth occurred (solid line, dotted lines denote earliest/latest start/end of growing season); (b) annual stem increments in 2019–2022 in Boubín and in 2016–2022 in Žofín; (c) the proportion of growing days, days with shrinkage, and days with stem tissue refiling; (d) daily annual increments and (e) daily shrinkage. Boxplots show 25th and 75th percentiles, the central line represents the median, the whiskers show maximum and minimum, and the dots the outliers. Individual species are color-coded (consistently through the manuscript).

(Fig. 3e–h). We observed this contrast mainly between broad-leaved *F. sylvatica* and coniferous *P. abies*. While *F. sylvatica* exhibited a higher proportion of growing days with smaller daily increments, *P. abies* showed the opposite growth pattern – higher increment on fewer growing days (Fig. 5; see also Etzold *et al.*, 2022; Salomón *et al.*, 2022). This demonstrates a species-specific nonlinearity in the climate–growth relationship of individual tree species (Wilmking *et al.*, 2020). Also, the effects of individual climatic variables on daily radial increments were in general stronger for *P. abies* and weaker for *F. sylvatica* (Fig. 3).

The different growth inhibition of the species investigated here is likely given by their different iso-anisohydricity. The isohydric *P. abies* closes its stomata and inhibits metabolism much earlier than the anisohydric *F. sylvatica* (Ulrich & Grossiord, 2023). Higher growth plasticity then allows *P. abies* to better time its growth to suitable conditions (Martinez del Castillo *et al.*, 2018; Zweifel *et al.*, 2021). On the other hand, earlier metabolic inhibition due to solar irradiance may cause *P. abies* to be more sensitive to other external factors, for example bark beetle attacks (Netherer *et al.*, 2024). By contrast, the lower growth plasticity of *F. sylvatica* (when compared with *P. abies*) to higher SDSI, RDR and VPD results in a higher resistance to drought from the perspective of tree growth (Ježík *et al.*, 2021; Kašpar *et al.*, 2021). In the long-term though, a lower growth plasticity could lead to negative growth trends (Martinez del Castillo *et al.*, 2022). Under a changing climate, these species-specific responses may alter the growth dynamics and timing of individual tree species, ultimately resulting in altered long-term growth trends (Kašpar *et al.*, 2024).

Model calibrations and the metrics used

In accordance with other studies (e.g. Peters *et al.*, 2021; Zweifel *et al.*, 2021; Tumajer *et al.*, 2022), VPD was identified as a strong predictor with significant physiological relevance. However, solar irradiance correlates with most climate metrics (Tsiropoula, 2003;

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Strahler, 2010), as evidenced by the high correlation between RDR and VPD (Fig. S9). Due to the mutual correlation of VPD with SDSI and RDR (Fig. S9), the effects of individual metrics on tree growth are hard to separate (Oogathoo *et al.*, 2020). Nevertheless, performed mediation analysis addressed this issue (Fig. 4; Table 2).

Suggested by the data on SWP measured at both sites, trees in our study were not stressed by the availability of soil water (Fig. S8). Excluding this effect, which many studies have shown to be strong (e.g. Babst *et al.*, 2019; Peters *et al.*, 2021; Zweifel *et al.*, 2021; Salomón *et al.*, 2022), allowed us to amplify the effects of atmospheric conditions on tree growth. However, this fact makes our results specific and they cannot be applied to dryer sites without caution.

We used climate predictors of only a short time scale (1 d, and in case of precipitation 2-d totals). Predictors capturing legacy effects, either intra- or inter-seasonal, were, therefore, not included. Despite this, our models were able to explain *c.* 49–68% of the variability in the probability of tree radial growth and 25–38% of the variability in the daily stem increment, which is comparable to other studies (van der Maaten *et al.*, 2018; Zweifel *et al.*, 2021; Tumajer *et al.*, 2022). We also observed a high variability captured by fixed predictors, suggesting large individual variability remaining (Tables 1, 2; Tumajer *et al.*, 2022). We also attempted to separate out the effect of precipitation on our results by conducting a separate analysis, excluding all days with precipitation (> 0.1). The results of this separate analysis remained consistent with the results given above (data not shown).

The E-OBS data used in this study well correlated with our *in situ* data, with the lowest correlation for precipitation (Fig. S4), and our results showed similar trends at both sites regardless of the climate data source (data not shown). To some extent, therefore, these results also indirectly indicate the lower effects of local extreme precipitation events (thunderstorms), which are poorly captured in gridded data.

It is important to note that we simulated solar irradiance of the visible part of tree crowns. A limitation of our study is that this method is only suitable for canopy trees (due to the 2.5D raster approach and insufficient simulation of diffuse and reflected light), and highly suppressed or sub-canopy trees may exhibit different growth responses.

Conclusions

This study leverages an extensive dataset of dendrometer data from two forests characterized by perennially adequate water for tree growth. Under such conditions, those atmospheric conditions that had, significant effects on solar irradiance controlled the growth of canopy trees. Specifically, intense solar irradiation demonstrated a minimal influence on the probability of growth on any given day, but markedly constrained the daily stem increment. This is attributable to the increasing VPD, leaf temperature, and transpiration rates with increasing solar irradiance, which collectively lower turgor and directly limit cell division in the cambial zone. Elevated rates of daily solar irradiance reduce the time-frame available for stem rehydration and tree growth. Consequently, tree growth was found to be maximized on long days of lower solar intensity. However, the optimal level of solar irradiance for tree growth appears to be species-specific, with conifers (*P. abies*) exhibiting higher plasticity to atmospheric conditions than broadleaves (*F. sylvatica*) at a daily scale. When comparing data from multiple growing seasons, *F. sylvatica* exhibited superior growth to *P. abies* only at higher elevation sites. *Abies alba* showed a comparable response to solar irradiance in terms of stem increment, yet it paralleled *F. sylvatica* in the number of growth days and achieved the highest annual stem increments.

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Competing interests

None declared.

Author contributions

JK and KK designed the project. JK collected and analyzed dendrometer data. MK analyzed LiDAR data. JK performed the statistical analysis. JK and KK wrote the manuscript.

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Data availability

All the data and materials that support the findings of this study are available at doi: 10.5281/zenodo.11127186.

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Supporting Information

Additional Supporting Information may be found online in the Supporting Information section at the end of the article.

Fig. S1 Map of the study areas.

Fig. S2 Morphometric characteristics of the studied trees.

Fig. S3 Methodology to eliminate annual trends in climate data.

Fig. S4 Comparison of selected daily climate metrics from the E-OBS database with *in situ* measured data.

Fig. S5 Violin plots showing the distribution of specific climate metrics on growing season days.

Fig. S6 The effect of tree size on the effect of solar irradiance.

Fig. S7 The timing and intensity of tree growth at both sites.

Fig. S8 Data of the volumetric soil moisture and soil water potential in Boubín and Žofín sites.

Fig. S9 The relationship of vapor pressure deficit and the relative daily irradiation.

Methods S1 R script used to calculate the solar irradiance.

Table S1 Dataset characteristics.

Table S2 Model formulas used in the study.

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